Likelihood: from first principles, v10

The surface $f(x, \theta)$ and its two readings

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1 One function, two readings

Fix a parametric family $\{f(\,\cdot\,;\theta) : \theta \in \Theta\}$ on a measurable space $(\mathcal{X}, \mathcal{A})$. Here $f(x, \theta)$ is a function of *two* variables: the data point $x \in \mathcal{X}$ and the parameter $\theta \in \Theta$.

What is fixed	What varies	Name	Interpretation
θ	x	Density $f(x;\theta)$	How probability is spread over outcomes
x (observed)	θ	Likelihood $\mathcal{L}(\theta; x)$	How compatible each θ is with the data

Same function — two perspectives. The surface $z = f(x, \theta)$ is a single mathematical object. Slicing it parallel to the x-axis gives densities. Slicing it parallel to the θ-axis gives likelihoods.

2 The Gaussian case: the surface in full

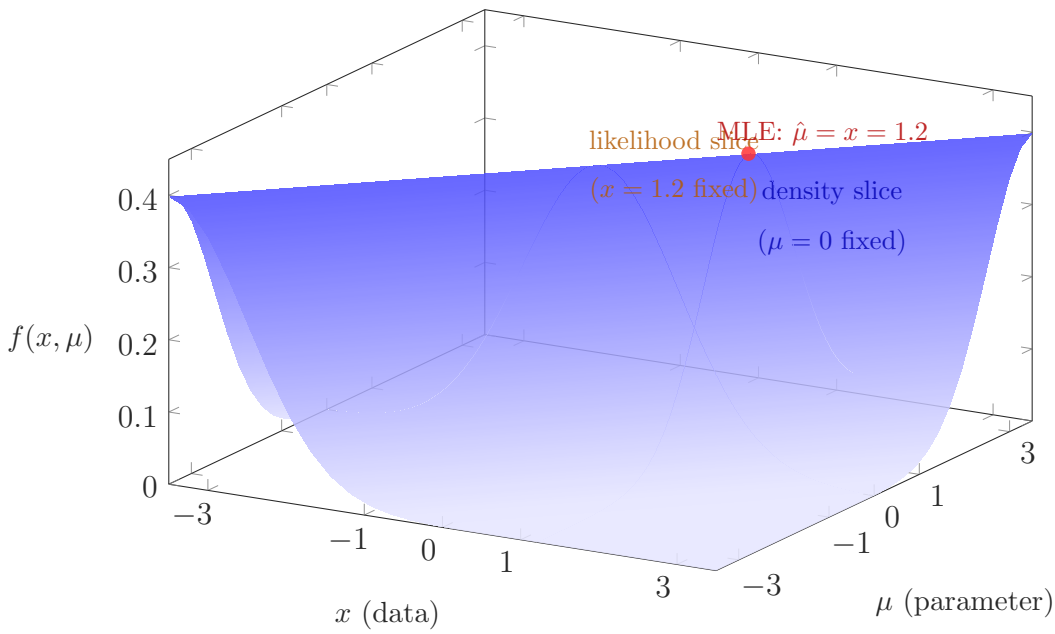
2.1 Setup

$\mathcal{X} = \mathbb{R}$, parameter $\theta = \mu \in \mathbb{R}$ (mean), known $\sigma = 1$:

$$f(x, \mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2}\right).$$

This is a function of two real variables. We can plot it as a surface in 3D with axes x (data), μ (parameter), f (height).

2.2 The 3D surface



2.3 Reading the surface: three key observations

Observation 1. Every horizontal slice (fixed μ, varying x) is a Gaussian density curve. They all have the same shape — just shifted left or right. Each integrates to 1: $\int_{-\infty}^{\infty} f(x, \mu)\, dx =$

1.

Observation 2 (corrected). Every vertical slice (fixed observed x, varying μ) is the likelihood function $\mathcal{L}(\mu) = f(x;\mu)$. For the Gaussian specifically, $f(x,\mu) = f(\mu,x)$: the function depends only on $(x-\mu)^2$, so it is *symmetric* in x and μ. This means the likelihood slice has *exactly the same bell shape* as the density slice, and integrating over μ gives:

$$\int_{-\infty}^{\infty} \mathcal{L}(\mu)\, d\mu = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(x-\mu)^2/2}\, d\mu = 1.$$

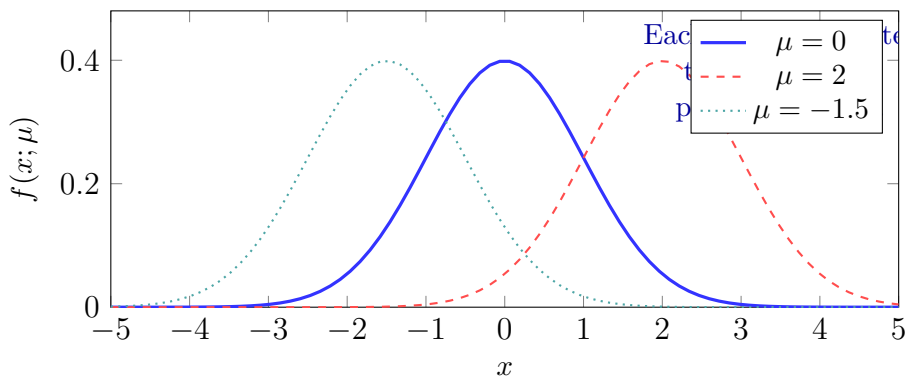
(Substitute $u = \mu - x$.) So for the Gaussian, the likelihood slice also integrates to 1 over μ. This is a special structural property of the Gaussian arising from its symmetry in $(x-\mu)$, not a general fact.

For a general parametric family the integral $\int_\Theta \mathcal{L}(\theta;x)\, d\theta$ can take any positive value and has no statistical interpretation. We cannot treat likelihood as a density over θ (that would require a prior — the Bayesian approach). The distinction between density and likelihood is *conceptual*, not about the value of any integral: density is what we integrate over $\mathcal{X}$; likelihood is what we maximise over Θ.

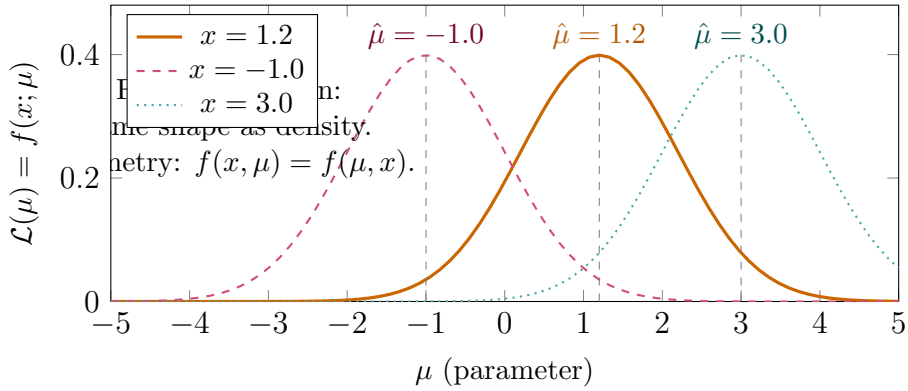
Observation 3. The ridge of the surface (the highest point of each likelihood slice) follows the diagonal $\mu = x$. So for any observed x, the MLE is $\hat{\mu} = x$. This is the crest running diagonally across the surface.

3 Two slices in 2D: density vs. likelihood

3.1 Density slices: fixed μ, varying x



3.2 Likelihood slices: fixed x, varying μ

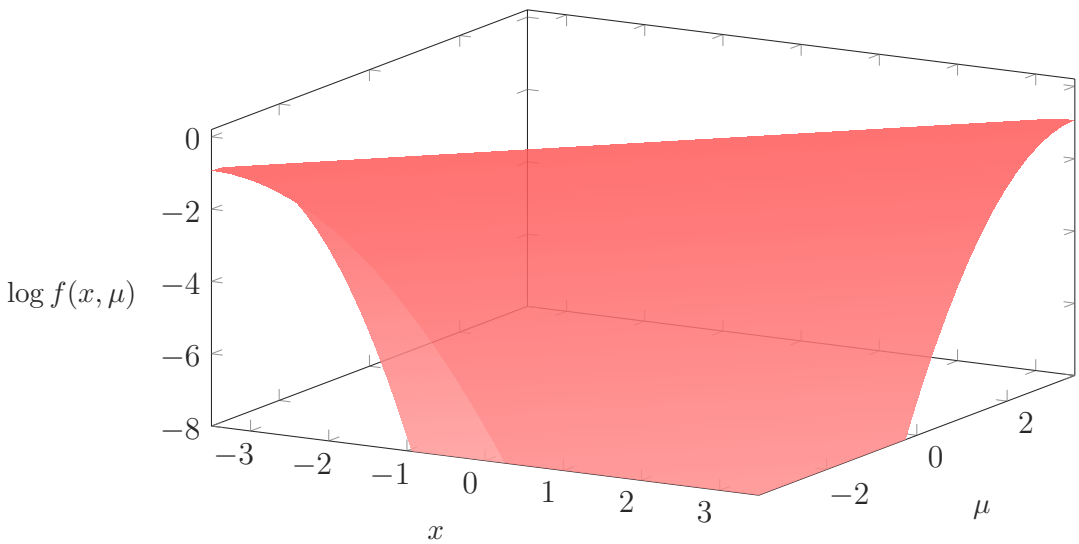


Key point: each likelihood curve peaks exactly at $\hat{\mu} = x$ (the vertical dashed lines). The MLE is the observed value — trivially, with one observation.

4 The log-likelihood surface

Working with $\log f(x, \mu)$ is more convenient. For the Gaussian:

$$\log f(x, \mu) = -\frac{(x-\mu)^2}{2} - \frac{1}{2}\log(2\pi).$$



Remark (The log-likelihood is a paraboloid). $\log f(x, \mu) = -\frac{1}{2}(x-\mu)^2 + \text{const}$ is a downward paraboloid in (x, μ) space, symmetric about the diagonal $x = \mu$. Its ridge is the diagonal: for each fixed x, the maximum over μ is at $\mu = x$. This is the geometric reason the Gaussian MLE is the sample mean.

5 The Bernoulli case: two separate parametrizations

Remark (Two fences live in different spaces). The triangular fences and the sigmoidal fences cannot share the same horizontal axis. Their supports are fundamentally different:

- Triangular fences: parameter $p \in (0, 1)$ — the probability itself, bounded.

- Sigmoidal fences: parameter $\eta \in \mathbb{R}$ — the linear predictor, unbounded.

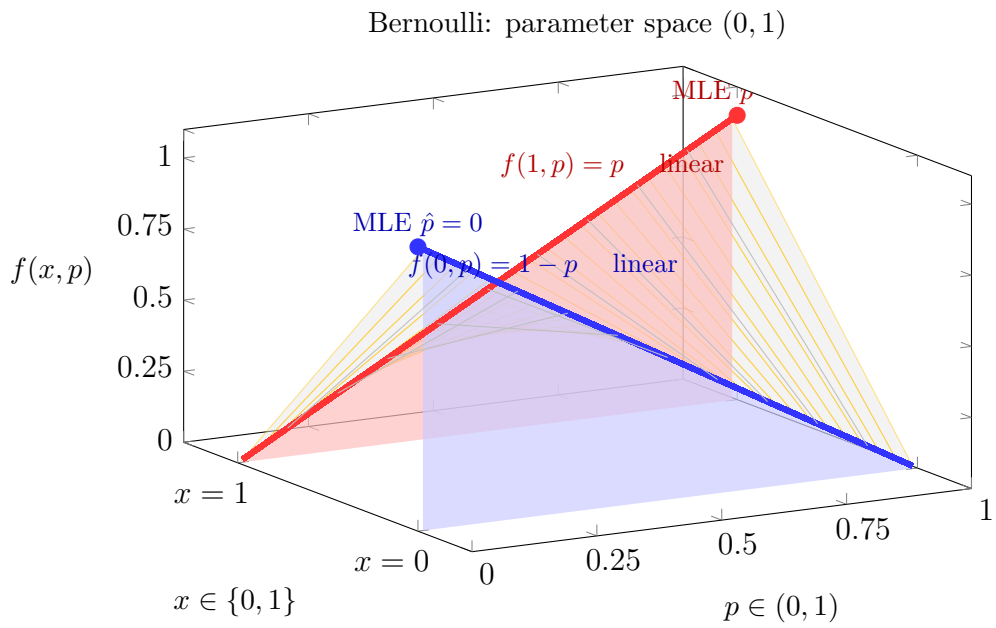
They are two *parametrizations of the same model*, connected by the logit bijection:

$$\eta = \operatorname{logit}(p) = \log \frac{p}{1-p}, \qquad p = \operatorname{expit}(\eta) = \frac{1}{1+e^{-\eta}}.$$

No linear rescaling of one axis produces the other. Each deserves its own figure.

5.1 Figure 1: p-parametrization — triangular fences

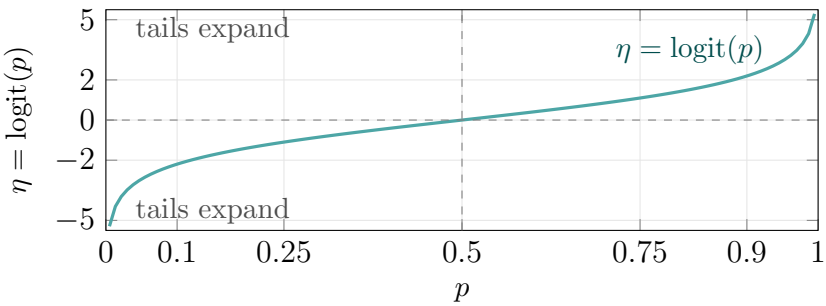
$f(x,p) = p^x(1-p)^{1-x}$, parameter $p \in (0,1)$.



The fences are *linear* (triangular) because $f(1,p) = p$ and $f(0,p) = 1-p$ are linear in p. The MLE has no interior maximum: it hits the boundary $\hat{p} \in \{0,1\}$.
Remark (Geometry: hyperbolic paraboloid). The connecting surface is $z(p,y) = 1 - p - y + 2yp$, a bilinear form, i.e. a **hyperbolic paraboloid** (saddle surface). It is a ruled surface: both the vertical grey bars (fixed p, vary y) and the lines of constant y (fixed y, vary p) are straight lines lying exactly on the surface. `pgfplots` with `samples y=2` renders this exactly — no approximation. The interpolation $y \in [0,1]$ is a visual aid only; $y \in \{0,1\}$ in the probability model.

5.2 Reparametrization bridge

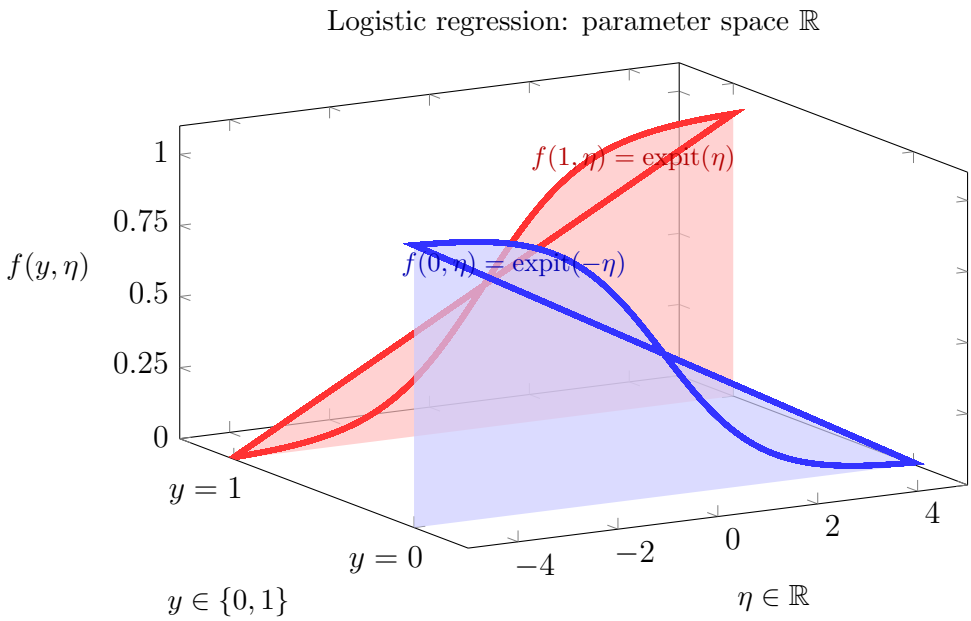
The logit map $\eta = \log \frac{p}{1-p}$ is a bijection $(0,1) \to \mathbb{R}$. It stretches the bounded p-axis into the unbounded η-axis, compressing the centre ($p \approx 0.5$, $\eta \approx 0$) and expanding the tails:



Under this map the *linear* fence $f(1, p) = p$ becomes the *sigmoidal* fence $f(1, \eta) = \text{expit}(\eta)$, and the linear fence $f(0, p) = 1 - p$ becomes $f(0, \eta) = \text{expit}(-\eta)$. This is not a new model — it is the same Bernoulli pmf, expressed in a different coordinate.

5.3 Figure 2: η-parametrization — sigmoidal fences

$f(y, \eta) = \text{expit}(\eta)^y \, \text{expit}(-\eta)^{1-y}$, parameter $\eta \in \mathbb{R}$. This is the likelihood surface of logistic regression (one observation, varying the linear predictor).



The fences are now *sigmoidal* because expit is the inverse logit. The MLE has no boundary problem: with n observations the log-likelihood $\sum_i [y_i \eta_i - \log(1 + e^{\eta_i})]$ is strictly concave in $\boldsymbol{\beta}$, so the score equation has a unique interior solution.

Remark (Geometry: ruled but not saddle). The connecting surface is $z(\eta, y) = y \, \text{expit}(\eta) + (1 - y) \, \text{expit}(-\eta)$. It is again a ruled surface (linear in y for every fixed η), but *not* a hyperbolic paraboloid: the η-direction curves sigmoidally, not linearly. As in Figure 1, `pgfplots` renders it exactly (linear in y, arbitrary in η, so `samples y=2` suffices).

6 What the surface reveals about MLE geometry

Reading the surface $f(x, \theta)$ as a function of two variables gives a unified picture:

1. **Density slices** (θ fixed, x varies): cross-sections parallel to the x-axis. Each must integrate (or sum) to 1 over $\mathcal{X}$.

2. **Likelihood slices** (x fixed, θ varies): cross-sections parallel to the θ-axis. Their heights are not normalised. The *shape* encodes how much each θ is supported by the observed x.
3. **The ridge** of the surface (maximum over θ for each fixed x): this is the MLE as a function of x. For the Gaussian it is the diagonal $\hat{\mu}(x) = x$.
4. **Curvature of the likelihood slice** at $\hat{\theta}$: this is related to Fisher information. A sharply curved (narrow) peak means high information — the data localise θ well. A flat peak means low information.

The next step (v3): aggregate n independent observations. The log-likelihood surface becomes a sum, the ridge sharpens, and the MLE converges to the truth with variance $1/(n\,\mathcal{I}(\theta))$.